

Experiment No. 21

Write a Prolog Program to print particular bird can fly or not. Incorporate required queries.

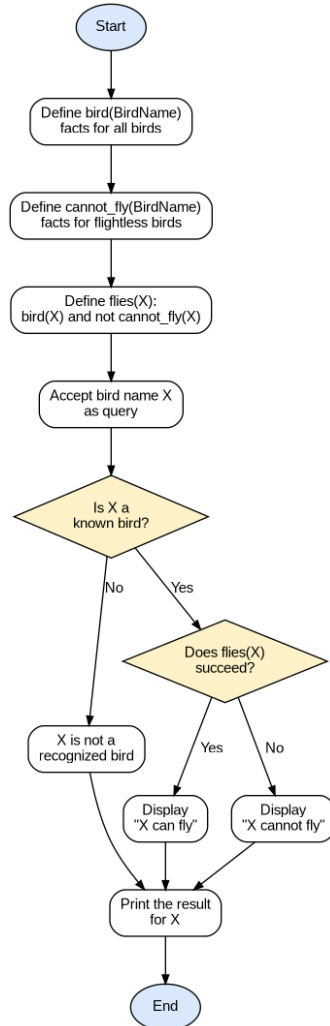
Aim

To write a Prolog program that determines whether a particular bird can fly or not, using facts and rules, and to incorporate the required queries to test the program for different birds.

Algorithm

1. Start.
2. Define facts `bird(BirdName)` for the set of known birds.
3. Define facts `cannot_fly(BirdName)` for birds that are unable to fly, such as penguin and ostrich.
4. Define the rule `flies(X)` which succeeds if X is a bird and X is not present in the `cannot_fly` facts.
5. Define the rule `check_flight(X)` that tests `flies(X)` for a given bird X.
6. If `flies(X)` succeeds, display the message "X can fly".
7. If X is a known bird but `flies(X)` fails, display the message "X cannot fly".
8. If X is not present in the bird facts at all, display the message "X is not a recognized bird".
9. Accept a bird name as a query to `check_flight`.
10. Repeat the check for multiple birds using separate queries.
11. Stop.

Flowchart



Program

bird(sparrow).

bird(eagle).

bird(pigeon).

bird(penguin).

bird(ostrich).

bird(crow).

cannot_fly(penguin).

cannot_fly(ostrich).

flies(X) :-

bird(X),

\+ cannot_fly(X).

```
check_flight(X) :-  
  ( flies(X) ->  
    format("~w can fly~n", [X])  
  ; bird(X) ->  
    format("~w cannot fly~n", [X])  
  ; format("~w is not a recognized bird~n", [X])  
  ).
```

```
main :-  
  check_flight(sparrow),  
  check_flight(penguin),  
  check_flight(ostrich),  
  check_flight(eagle),  
  check_flight(cat).
```

Result

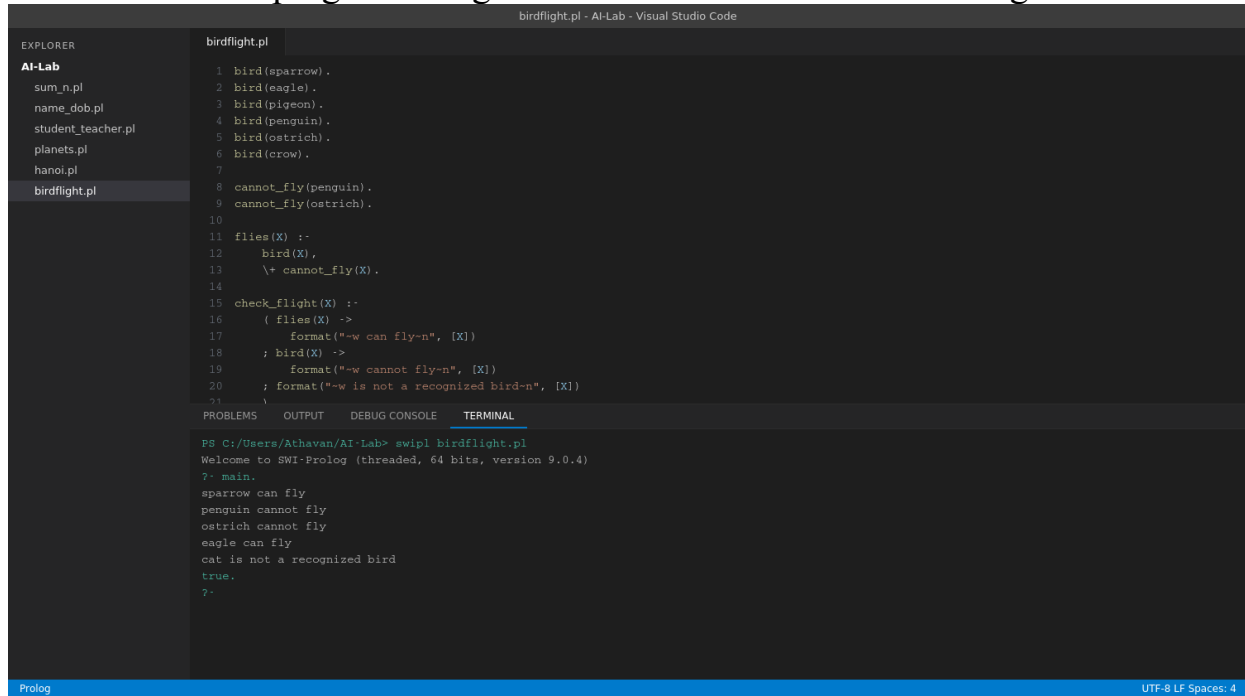
The program was executed successfully and the output was verified as correct.

?- main.

```
sparrow can fly  
penguin cannot fly  
ostrich cannot fly  
eagle can fly  
cat is not a recognized bird  
true.
```

Sample Output — Running in VS Code

Screenshot of the program being run in the Visual Studio Code integrated terminal:



The screenshot shows the Visual Studio Code interface with a file explorer on the left, a code editor in the center, and a terminal at the bottom. The file explorer shows a project named 'AI-Lab' with several files, including 'birdflight.pl'. The code editor displays the contents of 'birdflight.pl', which is a Prolog program. The terminal shows the output of running the program, which includes a welcome message and a list of birds that can fly, cannot fly, and are not recognized.

```
birdflight.pl - AI-Lab - Visual Studio Code

EXPLORER
AI-Lab
  sum_n.pl
  name_dob.pl
  student_teacher.pl
  planets.pl
  hanoi.pl
  birdflight.pl

1 bird(sparrow).
2 bird(eagle).
3 bird(pigeon).
4 bird(penguin).
5 bird(ostrich).
6 bird(crow).
7
8 cannot_fly(penguin).
9 cannot_fly(ostrich).
10
11 flies(X) :-
12     bird(X),
13     \+ cannot_fly(X).
14
15 check_flight(X) :-
16     ( flies(X) ->
17         format("~w can fly-n", [X])
18     ; bird(X) ->
19         format("~w cannot fly-n", [X])
20     ; format("~w is not a recognized bird-n", [X])
21     )
22
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL

PS C:/Users/Athavan/AI-Lab> swipl birdflight.pl
Welcome to SWI-Prolog (threaded, 64 bits, version 9.0.4)
?- main.
sparrow can fly
penguin cannot fly
ostrich cannot fly
eagle can fly
cat is not a recognized bird
true.
?-
```

Prolog UTF-8 LF Spaces: 4